

Computer Organization and Assembly Language



Lab Manual 07

Objectives:

1. Revision of reading input and storing in memory using DOS interrupts.
2. Understanding conversion of binary strings to numeric registers.
3. Learning various bitwise operations: ROL, ROR, SHL, SHR, AND, OR, XOR.
4. Printing numbers in binary format using assembly.
5. Understanding conditional operations based on specific bits.

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Task:

Write an assembly program that:

1. Reads a **binary number** from the keyboard (up to 16 bits).
 - **Note:** Users can enter fewer than 16 bits; the **Enter key** acts as the end of input.
 - Validate that only '0' and '1' are accepted.
2. Converts the entered binary into a **16-bit register value**.
 - **Approach options:**
 1. **Direct conversion:** Convert each bit to the register as it is entered.
 2. **Buffer method:** Store all input bits in a buffer first, then process the buffer to form the 16-bit value.
3. Performs the following **bitwise operations**:
 - If **LSB = 0**, perform **ROR 4**, else **ROL 4**.
 - If **MSB = 0**, perform **SHR 1**, else **SHL 1**.
 - Clear **bit 5** and set **bit 1**.
 - XOR the **high nibble (bits 15–12)** and **low nibble (bits 3–0)** with 0xF.
4. Prints the final 16-bit value in **binary**:
 - If **bit 3 = 0**, print **MSB → LSB**.
 - If **bit 3 = 1**, print **LSB → MSB**.

Test cases:

Input: 1011110 (7 bits)

Output: 0011000000000101

Input: 0

Output: 1011000000001111

Input: 11111111 (8 bits)

Output: 1111011111010101